

# Abstract – EJB Full-Stack Application: Semiconductor Inventory Management System

**Title:** Semiconductor Inventory Management System

## **Overview:**

This project aims to develop a full-stack Enterprise JavaBeans (EJB) application for managing semiconductor inventory, including components such as transistors, diodes, wafers, and integrated circuits (ICs). The system will streamline warehouse operations by providing CRUD functionalities for components and suppliers, recording import/export transactions, and generating real-time stock reports with a user-friendly interface.

## **Database Design:**

The application will be backed by a relational database with at least three core tables:

**Components:** Stores details of semiconductor components (name, category, quantity, reorder level).

**Suppliers:** Maintains supplier information.

**Transactions:** Records all import/export operations with timestamps and references to components and suppliers.

## **EJB Implementation:**

**Stateless Session Beans:** Handle business logic for managing components, suppliers, and transactions.

**Message-Driven Bean (MDB):** Listens to a JMS queue to process asynchronous events. For example, when a component's stock falls below its reorder threshold, a message is sent to the queue, and the MDB triggers an automated alert or generates a restocking request.

## **User Interface:**

A web-based UI will be developed to provide:

A dashboard with charts summarizing current inventory and monthly transaction trends.

Forms for managing components, suppliers, and transactions.

Visual alerts for low-stock components.

## **Innovation and Practicality:**

The project mimics a real-world **ERP-like warehouse management system** tailored for semiconductor components. By combining synchronous CRUD operations with asynchronous stock monitoring, the system demonstrates not only technical compliance with EJB standards but also high practical relevance.

## **Expected Outcome:**

A deployable EAR application with an integrated EJB back end, relational database, MDB-based asynchronous notification system, and a clean, user-friendly interface for managing semiconductor inventory.